

Total No. of Questions : 10]

SEAT No. :

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[5670]-140

B.E. (Mechanical)

OPERATION RESEARCH

(2012 Pattern) (Elective - II) (Semester - I)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8, Q9 or Q10.
- 2) Answers in One answer books.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) Discuss the various characteristics of Operation research. [4]

b) Reduce the following game by dominance and find the game value. [6]

		Player B			
		I	II	III	IV
Player A	I	3	2	4	0
	II	3	4	2	4
	III	4	2	4	0
	IV	0	4	0	8

OR

Q2) A company has four territories open and four salesmen available for an assignment. The territories are not equally rich in their sales potential. It is estimated that a typical salesman operating in each territory would bring in the following annual sales :

Territory	I	II	III	IV
Annual sales (Rs.)	1,26,000	1,05,000	84,000	63,000

P.T.O.

The four salesmen also differ in their ability. It is estimated that working under the same conditions, their yearly sales would be proportionality as follows :

Salesmen	A	B	C	D
Propositional	7	5	5	4

If the criterion is maximum expected total sales, the intuitive answer is to assign the best salesmen to the richest territory, the next best salesmen to the second richest and so on verify this answer by the assignment technique.[10]

- Q3)** a) Discuss in brief Hungarian method to solve the assignment problems.[4]
b) Find out the initial feasible solution by Vogel's Approximation Method (VAM). [6]

	W ₁	W ₂	W ₃	W ₄	Availability
F ₁	19	30	50	10	7
F ₂	70	30	40	60	9
F ₃	40	8	70	20	18
Requirement	5	8	7	14	

OR

- Q4)** A manufacturing firm produced a single product whose selling price is Rs. 16 per unit and the variable cost per unit are Rs. 12. If the fixed cost of the firm are estimated as Rs. 1,20,000/- find the break-even point in units, in Rupees and as a percentage of capacity if the firm has an estimated capacity of 50,000 units of the product. What is the margin of safety? [10]

- Q5)** a) Discuss in brief any two Inventory Control Techniques. [4]
b) An engineering is offered two types of material handling equipments A and B. A is priced at Rs. 5,00,000 including the cost of installation and the cost for the operation and maintenance are estimated as Rs. 60,000 for each of the first 5 years and increasing every year by Rs. 20,000 in the sixth and the subsequent year. Equipment B the same rated capacity

is priced at Rs. 2,50,000 including the cost of installation and the cost for the operation and maintenance are estimated as Rs. 1,20,000 for each of the first 5 years and increasing every year by Rs. 20,000 in the fifth year onwards. The company expected the return of 10% on all its investment. Neglect the scrap value of the equipment at the end of its economic life, determine which equipment the company should buy?[12]

OR

Q6) a) A manufacturer has to supply his customers with 600 units of his product per year. Shortage are not allowed and storage amount of 60 paise per unit per year. The set up cost per run is Rs. 80.00. Find out the

- i) Economic ordering quantity.
- ii) Minimum yearly average cost.
- iii) Minimum yearly total inventory cost. When the cost per unit item is Rs. 2/- per unit.
- iv) Optimum number of orders per year.
- v) The optimum period of supply per optimum order.

The increase in the order cost associated with ordering (1) 20% more than the EOQ. [8]

b) The following mortality rates have been observed for a certain type of fuse.

Week	1	2	3	4	5
% failing by the End of Week	5	15	35	75	100

There are 1000 fuses in use and it costs Rs. 5 to replace an individual fuse. If all fuses were replaced simultaneously it would cost Rs. 1.25 per fuse. It is proposed to replace all fuses at fixed intervals of time, whether or not they have burnt out and to continue replacing burnt out fuses as they fail. At what intervals the group replacement should be made? Also prove that this optimal policy is superior to the straight forward policy of replacing each fuse only when it fails. [8]

- Q7)** a) Find out the sequence that minimizes the total elapsed time required to complete the tasks on two machines. [10]

Operation	A	B	C	D	E	F	G	H	I
Machine 1	2	5	4	9	6	8	7	5	4
Machine 2	6	8	7	4	3	9	3	8	11

- b) Explain with the help of neat sketch a generalised queuing model. Also explain the Kendall's notation for representing queuing models? [6]

OR

- Q8)** a) A repair shop attended by a single mechanic has an average of four customers an hour who bring small appliances for repair. The mechanic inspects them for defects and takes six minutes an average. Arrivals are Poisson and service rate has the exponential distribution. [4]

You are required to

- What is the average number of customers in the system?
 - Find the average time spent by a customer in the shop including service.
- b) Six jobs are to be process on three machines. The processing time is as follows. Find the optimal schedule so that the total elapsed time is minimized. [12]

Job	J1	J2	J3	J4	J5	J6
Machine M1	10	3	5	4	2	1
Machine M2	2	4	6	3	1	2
Machine M3	8	6	7	9	7	7

Q9) a) Discuss CPM and PERT.

[6]

b) A bank has decided to modernize its office. The major elements of the project are as follows. **[12]**

Activity	Description	Predecessor Activity	Duration (Days)
A	Design New premises	---	14
B	Obtain tenders from contractors	A	4
C	Select the contractor	B	2
D	Arrange details with selected contractor	C	1
E	Decide which equipment is to be used	A	2
F	Arrange storage of equipment	E	3
G	Arrange disposal of other equipment	E	2
H	Order new equipment	E	4
I	Take delivery of new equipment	H,L	3
J	Renovations take place	K	12
K	Remove old equipment for storage or disposal	D,F,G	4
L	Cleaning after the contractor finished	J	2
M	Return old equipment for storage	H,L	2

i) Draw an arrow diagram for this project.

ii) Find out the critical path

iii) For each non-critical activity find out the total, free and independent float or Slacks.

OR

Q10) a) A project has the following data.

[12]

Task	1-2	1-3	1-4	2-5	2-6	3-6	4-7	5-7	6-7
Optimistic time	5	18	26	16	15	6	7	7	3
Pessimistic time	10	22	40	20	25	12	12	9	5
Most likely time	8	20	33	18	20	9	10	8	4

Determine the following :

- Construct the network and find out the expected task time and their variance.
- The earliest and the latest expected completion times of each event.
- The critical path.
- The probability of an event occurring at the expected completion date if the original scheduled time of completing the project is 41.5 weeks.
- The duration of the project that will have 95% chance of being completed.

Use following table

Probability	0.3085	0.3050	0.3015	0.2981	0.95
Normal distribution constant (Z)	-0.50	-0.51	-0.52	-0.53	1.64

b) Discuss in brief (Any Two).

[6]

- Monte-Carlo Simulation.
- Common Errors/Flaws in Network.
- Goal Programming.
- Crashing in the network.
